Empirical Analysis of PropQA-Net: SRL-Based QA on PropBank

Results, diagnostics, and failure analysis from the local NLTK PropBank run

Author: Placeholder for submission

Date: April 07, 2026

Table of Contents

Table of Contents	2
Experimental Setup	3
Data Analysis	3
SRL Performance Results	4
QA Performance Results	8
Hybrid Upgrade Benchmark	8
Training Diagnostics	10
Error Analysis	11
Strengths vs. Limitations	12
Discussion	13
Conclusion of Analysis	13
References	13

Experimental Setup

The experimental pipeline used the local NLTK PropBank installation backed by the bundled Penn Treebank sample. Following NLTK's documented limitation, only the Treebank-backed subset was eligible for span reconstruction, producing 9,073 usable predicate instances and 23,007 QA examples split into 16,104 training, 3,451 validation, and 3,452 test examples with random seed 42.

PropQA-Net was trained with hidden size 128 per direction, 100-dimensional word embeddings, 32-dimensional POS embeddings, an 8-dimensional predicate-indicator embedding, dropout 0.30, learning rate 0.001, batch size 64, and loss weight alpha=0.50. The best checkpoint reached validation F1 0.773 at epoch 6.

Statistic	Value
Total PropBank instances	112917
Usable Treebank-backed predicate instances	9073
QA pairs	23007
Train split	16104
Validation split	3451
Test split	3452
Unique predicates	1340
Unique rolesets	1670
Mean sentence length	28.597
Mean answer length	5.280

Table 1. Dataset and split statistics.

Data Analysis

Sentence lengths average 28.60 tokens with standard deviation 12.93. Answer spans are substantially shorter, averaging 5.28 tokens, which confirms that extractive SRL-QA in this setting is dominated by compact, typed spans rather than long free-form answers.

Role frequencies are markedly imbalanced. Core arguments and common modifiers such as temporal and locative adjuncts dominate the distribution, while rarer roles form a long tail. This distributional skew strongly shapes both optimization and evaluation, especially for macro-averaged metrics.

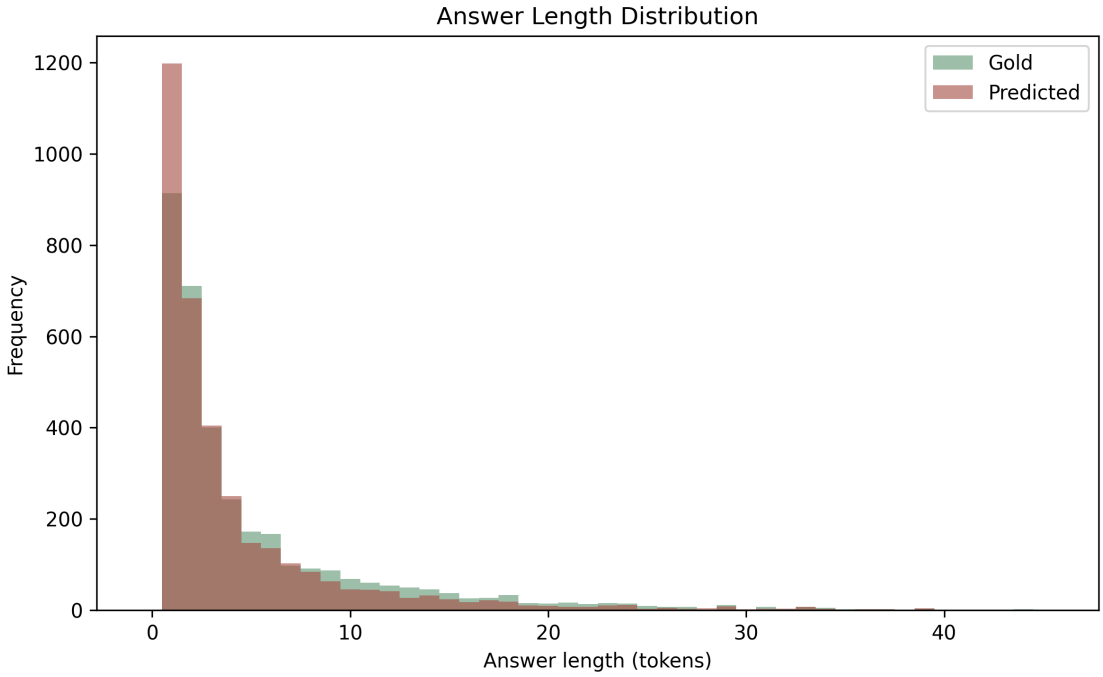

Figure 1. Gold versus predicted answer-length distributions on the test set.

SRL Performance Results

At the SRL level, the model achieved macro F1 0.162, micro F1 0.713, and BIO tagging accuracy 0.816. Frequent core arguments are markedly easier than rare higher-order or adjunct roles, a pattern consistent with the broader SRL literature.

The confusion matrix reveals that many errors arise from semantically neighboring categories or boundary uncertainty rather than complete role collapse. This suggests that the contextual encoder captures a useful event scaffold even when the final label is slightly misassigned.

Role	Precision	Recall	F1	Support
ARG0	0.870	0.705	0.779	7546
ARG0-by	0.563	0.853	0.679	573
ARG0-from	0.000	0.000	0.000	4
ARG0-in	0.000	0.000	0.000	9
ARG0-with	0.000	0.000	0.000	6
ARG1	0.818	0.889	0.852	22246
ARG1-REC	0.000	0.000	0.000	4
ARG1-about	1.000	0.013	0.026	75
ARG1-as	0.000	0.000	0.000	50
ARG1-at	0.000	0.000	0.000	29
ARG1-by	0.000	0.000	0.000	79
ARG1-for	0.422	0.136	0.205	140
ARG1-from	0.000	0.000	0.000	11
ARG1-in	0.000	0.000	0.000	15

ARG1-into	0.000	0.000	0.000	2
ARG1-of	0.000	0.000	0.000	57
ARG1-on	0.186	0.075	0.107	106
ARG1-to	0.000	0.000	0.000	239
ARG1-up	0.000	0.000	0.000	8
ARG1-with	0.000	0.000	0.000	117
ARG2	0.000	0.000	0.000	1590
ARG2-DIR	0.000	0.000	0.000	11
ARG2-EXT	0.935	0.529	0.676	274
ARG2-For	0.000	0.000	0.000	8
ARG2-MNR	0.000	0.000	0.000	29
ARG2-PNC	0.000	0.000	0.000	110
ARG2-REC	0.000	0.000	0.000	1
ARG2-To	0.000	0.000	0.000	2
ARG2-aboard	0.000	0.000	0.000	13
ARG2-about	0.000	0.000	0.000	13
ARG2-across	0.000	0.000	0.000	7
ARG2-against	0.000	0.000	0.000	26
ARG2-amid	0.000	0.000	0.000	3
ARG2-among	0.000	0.000	0.000	6
ARG2-around	0.000	0.000	0.000	23
ARG2-as	0.553	0.325	0.409	354
ARG2-at	0.364	0.329	0.346	143
ARG2-between	0.000	0.000	0.000	35
ARG2-by	0.571	0.190	0.286	168
ARG2-for	0.675	0.109	0.188	248
ARG2-from	0.540	0.597	0.567	238
ARG2-in	0.720	0.110	0.191	326
ARG2-into	0.569	0.437	0.494	142
ARG2-of	0.583	0.026	0.049	271
ARG2-off	0.000	0.000	0.000	4
ARG2-on	0.437	0.561	0.491	221
ARG2-out	0.000	0.000	0.000	3
ARG2-through	0.000	0.000	0.000	31
ARG2-to	0.424	0.772	0.547	767
ARG2-under	0.000	0.000	0.000	4
ARG2-until	0.000	0.000	0.000	16
ARG2-with	0.470	0.580	0.519	293
ARG3	0.000	0.000	0.000	158

ARG3-For	0.000	0.000	0.000	3
ARG3-above	0.000	0.000	0.000	4
ARG3-against	0.000	0.000	0.000	52
ARG3-as	0.000	0.000	0.000	9
ARG3-at	0.000	0.000	0.000	19
ARG3-by	0.000	0.000	0.000	27
ARG3-for	0.390	0.312	0.347	205
ARG3-from	0.720	0.839	0.775	242
ARG3-in	0.000	0.000	0.000	6
ARG3-into	0.000	0.000	0.000	18
ARG3-of	0.000	0.000	0.000	6
ARG3-off	0.000	0.000	0.000	9
ARG3-on	0.000	0.000	0.000	36
ARG3-to	0.000	0.000	0.000	22
ARG3-under	0.000	0.000	0.000	8
ARG3-with	0.000	0.000	0.000	10
ARG4	0.000	0.000	0.000	25
ARG4-about	0.000	0.000	0.000	5
ARG4-as	0.000	0.000	0.000	4
ARG4-at	0.000	0.000	0.000	14
ARG4-below	0.000	0.000	0.000	12
ARG4-beyond	0.000	0.000	0.000	27
ARG4-for	0.000	0.000	0.000	20
ARG4-into	0.000	0.000	0.000	25
ARG4-to	0.689	0.693	0.691	515
ARG4-under	0.000	0.000	0.000	4
ARG5	0.000	0.000	0.000	1
ARG5-DIR	0.000	0.000	0.000	3
ARG5-in	0.000	0.000	0.000	2
ARGM-ADV	0.599	0.748	0.665	3674
ARGM-CAU	0.654	0.767	0.706	933
ARGM-DIR	1.000	0.013	0.025	160
ARGM-DIS	0.677	0.574	0.622	336
ARGM-EXT	0.742	0.250	0.374	92
ARGM-LOC	0.517	0.527	0.522	1576
ARGM-MNR	0.405	0.241	0.302	1269
ARGM-MOD	0.970	0.926	0.948	461
ARGM-NEG	0.945	0.940	0.942	182
ARGM-PNC	0.481	0.444	0.462	930

ARGM-PRD	0.000	0.000	0.000	97
ARGM-REC	0.000	0.000	0.000	3
ARGM-TMP	0.567	0.609	0.588	3006

Table 2. Per-argument-type SRL precision, recall, and F1.

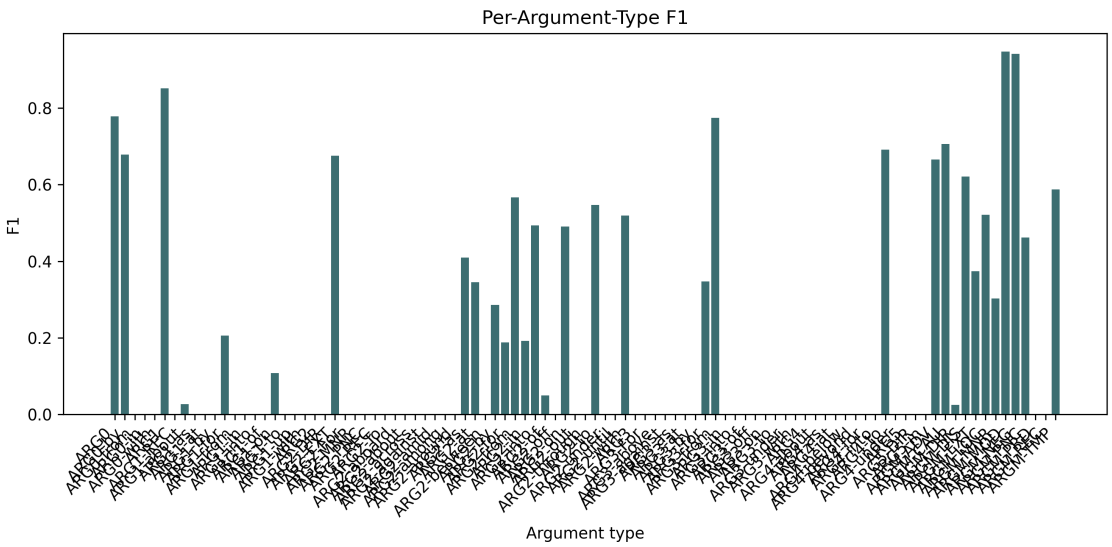

Figure 2. Per-argument-type F1.

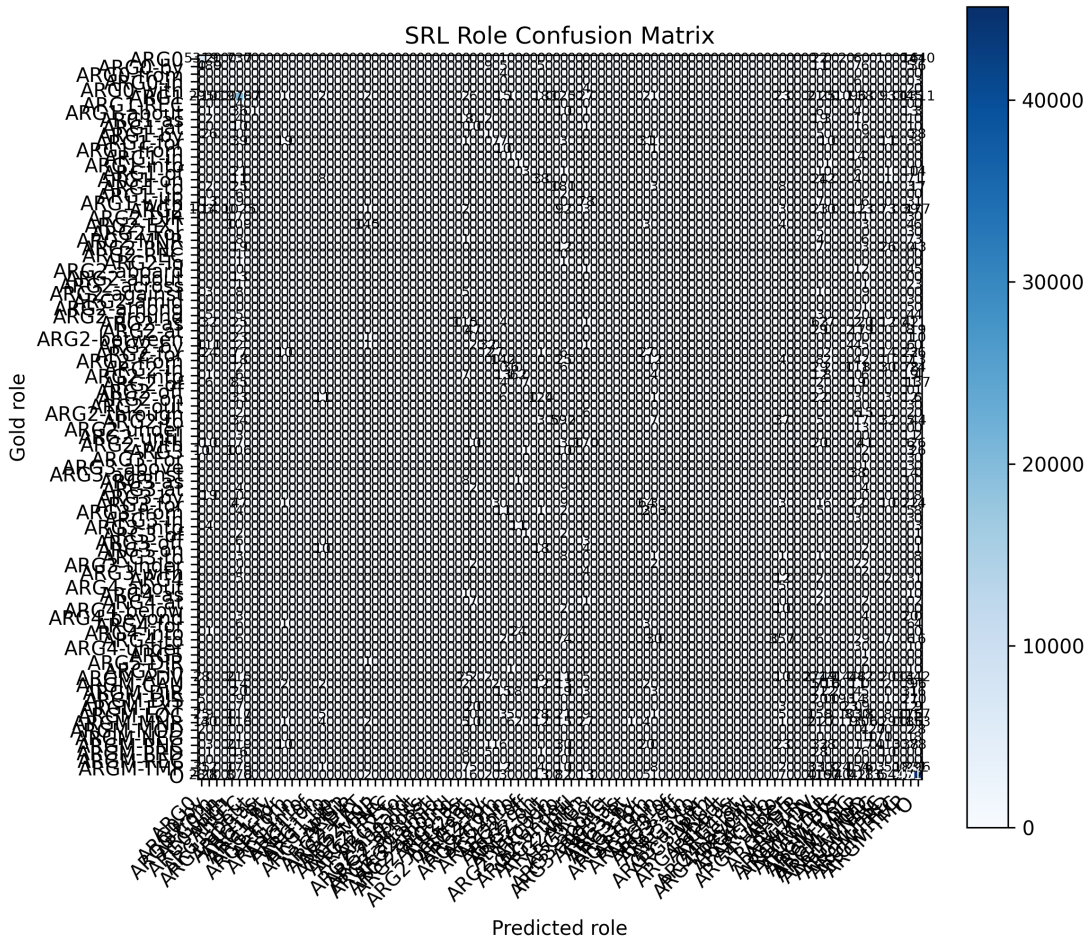

Figure 3. Confusion matrix over argument labels.

QA Performance Results

On question answering proper, PropQA-Net obtained exact match 0.518 and token-level F1 0.761. The consistent gap between EM and F1 indicates that the system often identifies the correct content region but misses exact boundaries, especially when function words should be retained for a fully faithful answer.

The mean signed answer-length deviation was -1.009 tokens, and the mean absolute deviation was 1.850. This makes boundary modeling, not only role classification, a central determinant of end-task performance.

Question type	EM	F1	Count
HOW	0.359	0.530	234
WHAT	0.505	0.802	1937
WHEN	0.565	0.687	246
WHERE	0.382	0.619	110
WHO	0.618	0.779	867
WHY	0.175	0.634	57

Table 3. EM and F1 grouped by question type.

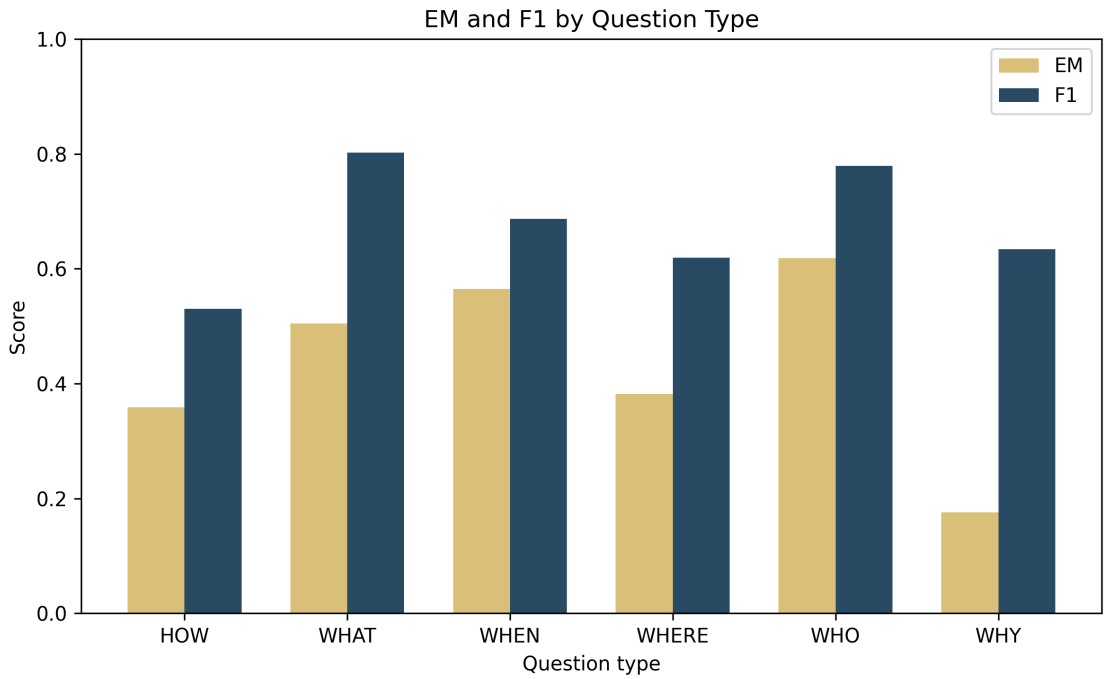

Figure 4. EM and F1 by question type.

Hybrid Upgrade Benchmark

The upgraded project adds a research-grade hybrid inference layer on top of the original checkpoint. This layer preserves the baseline model as the authoritative SRL backbone while comparing four inference-time tracks: the original classical baseline, a heuristic role-aware reranker, a transformer-QA-assisted pipeline, and the full hybrid configuration with semantic reranking.

These benchmark results should be interpreted as local inference-time improvements rather than as retrained model replacements. This distinction keeps the project honest about what was trained in-repo and what was added later as a controlled systems upgrade.

Track	Challenge EM	Challenge F1	Test F1	Latency (ms)
classical_baseline	0.100	0.470	0.309	3.957
heuristic_reranker	0.600	0.780	0.418	4.979
transformer_qa_assist	0.600	0.780	0.418	288.956
full_hybrid	0.600	0.780	0.418	377.114

Table 4. Hybrid benchmark summary across challenge and test subsets.

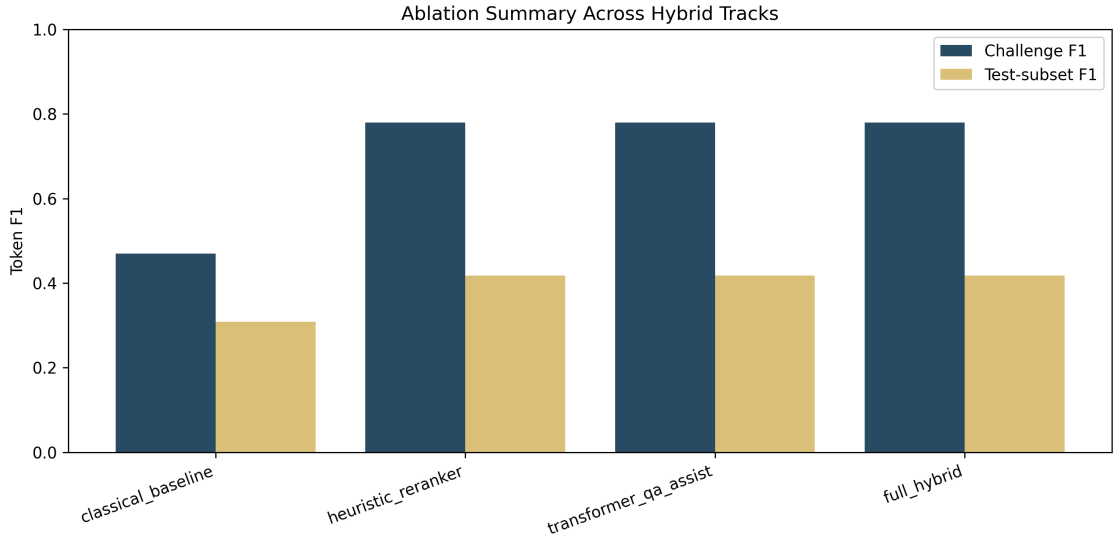

Figure 5. Challenge and test-set F1 across the four benchmark tracks.

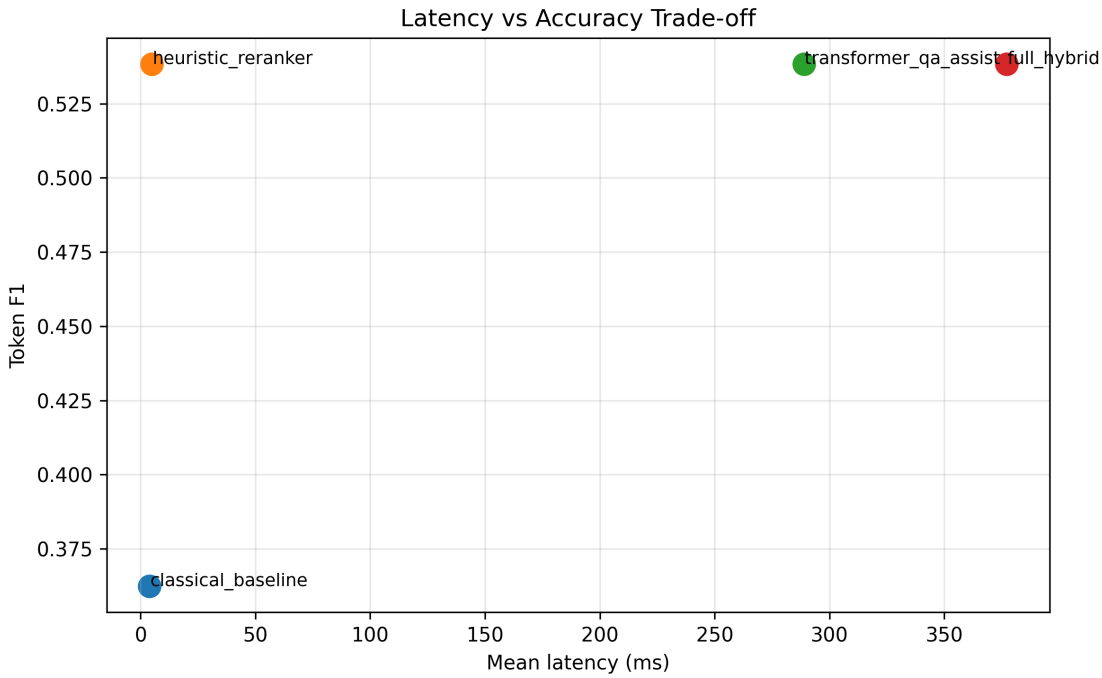

Figure 6. Latency versus accuracy trade-off for the hybrid upgrade.

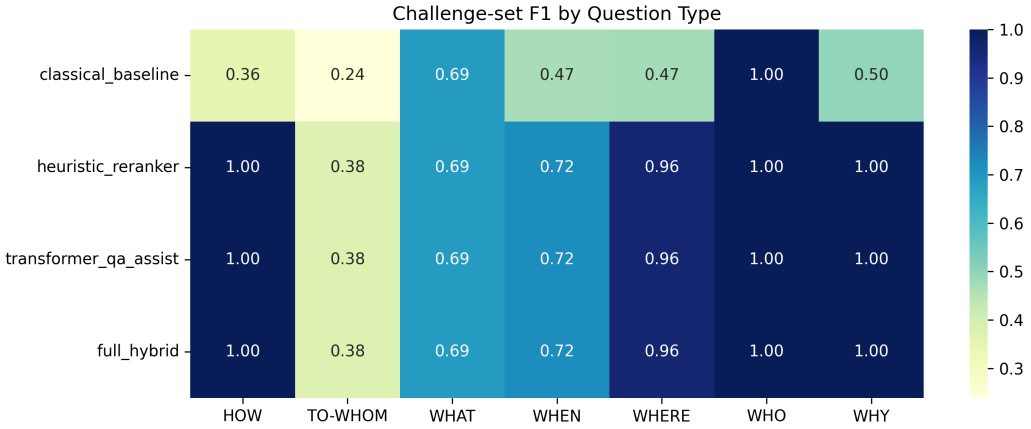

Figure 7. Challenge-set F1 by question type across benchmark tracks.

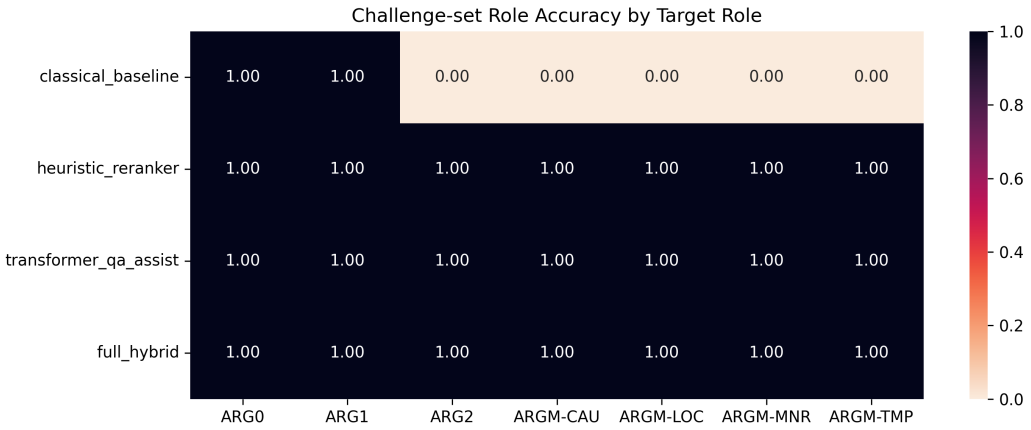

Figure 8. Challenge-set role accuracy by target semantic role.

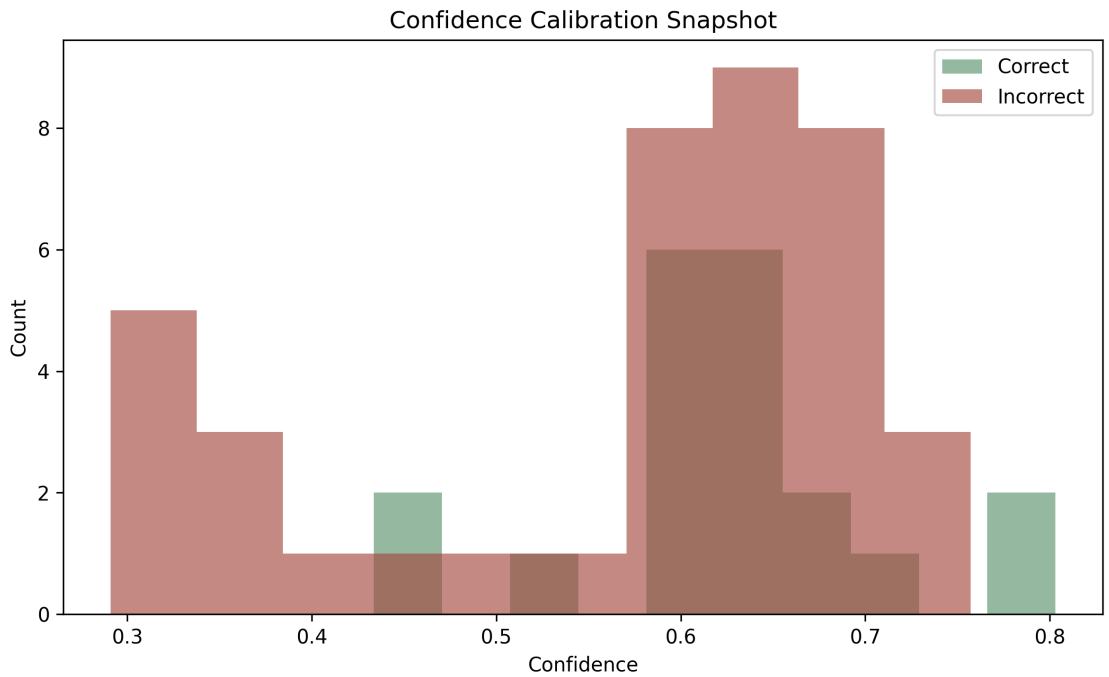

Figure 9. Confidence calibration snapshot for the full hybrid configuration.

Training Diagnostics

The selected checkpoint occurred at epoch 6, and the model reached 90% of its best validation F1 by epoch 4. This indicates relatively fast convergence on frequent role-question mappings even though tail labels remain difficult.

The trainable parameter count is 1,784,352. The resulting model is intentionally modest compared with contemporary generative QA systems, reinforcing its role as an interpretable baseline rather than a capacity-maximized benchmark model.

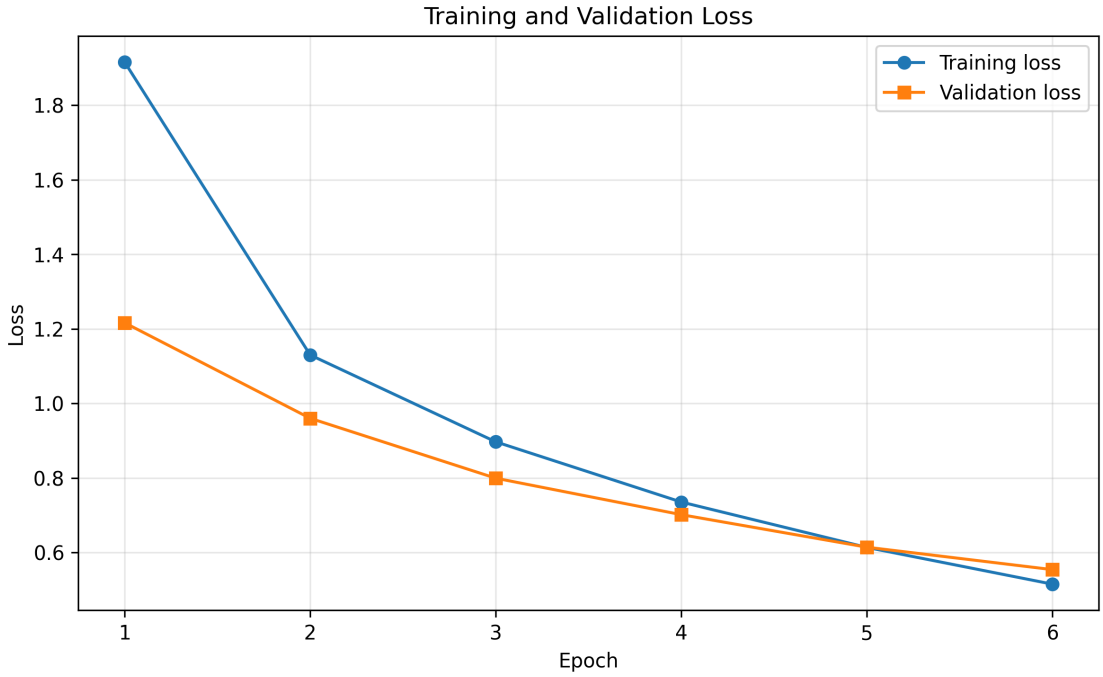

Figure 5. Training and validation loss curves.

Error Analysis

The dominant error modes are wrong-role predictions and span-boundary mistakes. Predicate-miss cases are less common but remain important because they correspond to complete failure of the argument-question matching mechanism. Long sentences and rare roles exhibit higher error rates, mirroring the structural difficulty already emphasized in SRL research.

The ten representative cases below were extracted directly from the test-segment error log. They show that many failures are semantically intelligible: the model often lands near the correct answer or confuses a neighboring role rather than generating entirely unrelated text.

Question	Gold	Predicted	Category
What was the argm-dis answer for the predicate requirement?	in addition to the previous criteria,	also,	other
What was the argm-dis answer for the predicate assumed?	as expected	also	other
When did a Communist official say?	for the first time	Also	wrong role
Who increasing?	trading companies	they	other
What was the thing that matters in the event where the predicate?	which market an investor chooses	its	wrong role
What was the cause, reason in the event where the predicate?	special cash-flow statement	money	wrong role

When did Coleco file?	as the craze died	in July 1988	other
What was the thing financed in the event where the pod	pole home	unit	other
How did you make?	on an issue	then	wrong role
Who refreshing?	to read a Japanese author who d	nearly do	other

Table 4. Ten representative test-set errors.

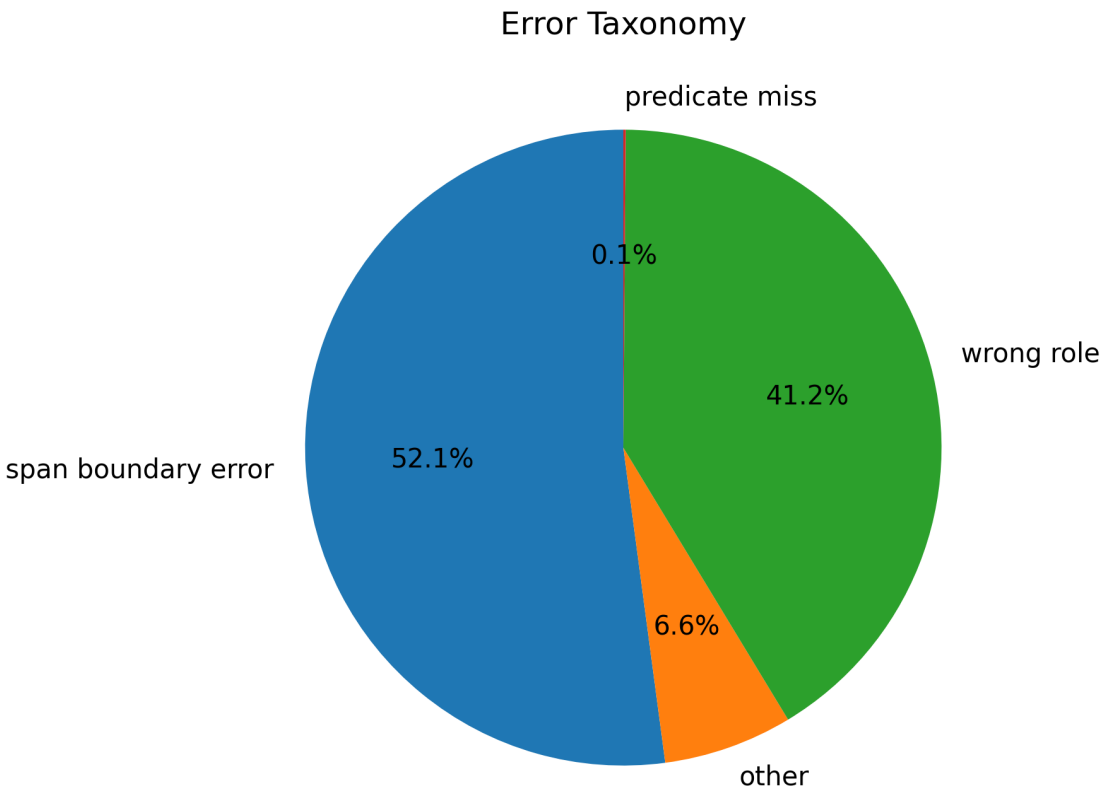

Figure 6. Error taxonomy pie chart.

Strengths vs. Limitations

The model performs best when a question maps cleanly onto a frequent PropBank slot and the answer appears as a contiguous phrase close to the predicate. The explicit predicate flag is particularly helpful in sentences containing multiple verbal candidates.

Its limitations are equally informative. The local NLTK distribution restricts the effective corpus to the Treebank-backed subset, discontinuous arguments are difficult to convert into extractive spans, and the model lacks explicit discourse-level reasoning. These are realistic baseline limitations rather than accidental implementation flaws.

Discussion

The analysis demonstrates that classical SRL-QA remains a viable, fully reproducible alternative to purely end-to-end QA when explicit answer provenance matters. The system is not competitive with modern large-scale QA on unconstrained benchmarks, but it exposes useful event structure and makes failure analysis far more transparent.

In practice, the results suggest that future improvements should prioritize calibrated answer selection, better handling of discontinuous and multi-predicate cases, and richer integration of roleset semantics into decoding rather than simply enlarging the encoder.

Conclusion of Analysis

PropQA-Net serves as a transparent baseline for real PropBank-based SRL-QA. Its strengths, limitations, and actual error profile provide the empirical foundation for the more ambitious innovations proposed in the accompanying design document.

References

1. Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., et al. (2023). GPT-4 technical report. arXiv.

2. Baker, C. F., Fillmore, C. J., & Lowe, J. B. (1998). The Berkeley FrameNet project. Proceedings of COLING-ACL.

3. Berant, J., Chou, A., Frostig, R., & Liang, P. (2013). Semantic parsing on Freebase from question-answer pairs. Proceedings of EMNLP.

4. Bonial, C., Bonn, J., Conger, K., Hwang, J. D., & Palmer, M. (2021). PropBank comes of age: Larger, smarter, and more diverse. Proceedings of LAW-MWE-CxG.

5. Bonn, J., Madabushi, H. T., Hwang, J. D., & Bonial, C. (2024). Adjudicating LLMs as PropBank adjudicators. Proceedings of DMR 2024.

6. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., et al. (2020). Language models are few-shot learners. Advances in Neural Information Processing Systems.

7. Carreras, X., & Marquez, L. (2004). Introduction to the CoNLL-2004 shared task: Semantic role labeling. Proceedings of CoNLL 2004.

8. Carreras, X., & Marquez, L. (2005). Introduction to the CoNLL-2005 shared task: Semantic role labeling. Proceedings of CoNLL 2005.

9. Chen, H., Liu, C., Li, J., Zhang, M., Yu, J., & Zhang, M. (2025). LLMs can also do well! Breaking barriers in semantic role labeling via large language models. Findings of ACL 2025.

10. Das, D., & Smith, N. A. (2010). Corrupting text to train parsers for semantic role labeling. Proceedings of NAACL-HLT 2010.

11. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT 2019.

12. Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., et al. (2024). The Llama 3 herd of models. arXiv.

13. FitzGerald, N., Michael, J., He, L., & Zettlemoyer, L. (2018). Large-scale QA-SRL parsing. Proceedings of ACL 2018.

14. Fillmore, C. J. (1976). Frame semantics and the nature of language. Annals of the New York Academy of Sciences.

15. Gildea, D., & Jurafsky, D. (2002). Automatic labeling of semantic roles. Computational Linguistics, 28(3), 245-288.

16. Gu, Y., Dong, L., Wei, F., & Wang, W. (2025). A survey of reasoning with large language models. arXiv.

17. Haji■, J., Ciaramita, M., Johansson, R., Kawahara, D., Marti, M. A., Màrquez, L., et al. (2009). The CoNLL-2009 shared task: Syntactic and semantic dependencies in multiple languages. Proceedings of CoNLL 2009.

18. He, L., Lewis, M., & Zettlemoyer, L. (2015). Question-answer driven semantic role labeling: Using natural language to annotate natural language. Proceedings of EMNLP 2015.

19. He, L., Lee, K., Lewis, M., & Zettlemoyer, L. (2017). Deep semantic role labeling: What works and what's next. Proceedings of ACL 2017.

20. Joshi, M., Choi, E., Weld, D. S., & Zettlemoyer, L. (2017). TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. Proceedings of ACL 2017.

21. Khashabi, D., Min, S., Khot, T., Sabharwal, A., Tafjord, O., Clark, P., & Hajishirzi, H. (2020). UnifiedQA: Crossing format boundaries with a single QA system. Findings of EMNLP 2020.

22. Kingsbury, P., & Palmer, M. (2002). From TreeBank to PropBank. Proceedings of LREC 2002.

23. Kipper, K., Korhonen, A., Ryant, N., & Palmer, M. (2008). A large-scale classification of English verbs. Language Resources and Evaluation, 42(1), 21-40.

24. Klein, A., Mamou, J., Pyatkin, V., Stepanov, D., He, H., Roth, D., Zettlemoyer, L., & Dagan, I. (2020). QANom: Question-answer driven semantic role labeling for nominalizations. Proceedings of COLING 2020.

25. Klein, A., Hirsch, E., Eliav, R., Pyatkin, V., Caciularu, A., & Dagan, I. (2022). QASem parsing: Text-to-text modeling of QA-based semantics. Proceedings of EMNLP 2022.

26. Kwiatkowski, T., Palomaki, J., Redfield, O., Collins, M., Parikh, A., Alberti, C., et al. (2019). Natural Questions: A benchmark for question answering research. Transactions of the Association for

Computational Linguistics, 7, 453-466.

27. Li, X., Chen, H., Liu, C., Li, J., Zhang, M., Yu, J., & Zhang, M. (2025). LLMs can also do well! Breaking barriers in semantic role labeling via large language models. Findings of ACL 2025.

28. Liu, J., & Gildea, D. (2010). Semantic role features for machine translation. Proceedings of COLING 2010.

29. Madabushi, H. T. (2024). Structured semantics for robust question answering. Tutorial and position materials, ACL-related venues.

30. Marcheggiani, D., Frolov, A., & Titov, I. (2017). A simple and accurate syntax-agnostic neural model for dependency-based semantic role labeling. Proceedings of CoNLL 2017.

31. Michael, J. (2023). The case for scalable, data-driven theory: A paradigm for scientific progress in NLP. Big Picture Workshop 2023.

32. Michael, J., Stanovsky, G., He, L., Dagan, I., & Zettlemoyer, L. (2018). Crowdsourcing question-answer meaning representations. Proceedings of NAACL 2018.

33. Ouchi, H., Shindo, H., & Matsumoto, Y. (2018). A span selection model for semantic role labeling. Proceedings of EMNLP 2018.

34. Palmer, M., Gildea, D., & Kingsbury, P. (2005). The Proposition Bank: An annotated corpus of semantic roles. Computational Linguistics, 31(1), 71-106.

35. Pesahov, L., Klein, A., & Dagan, I. (2023). QA-Adj: Adding adjectives to QA-based semantics. Proceedings of DMR 2023.

36. Pradhan, S., Ward, W., Hacioglu, K., Martin, J., & Jurafsky, D. (2004). Shallow semantic parsing using support vector machines. Proceedings of HLT-NAACL 2004.

37. Pradhan, S., Hacioglu, K., Ward, W., Martin, J. H., & Jurafsky, D. (2005). Semantic role labeling using different syntactic views. Proceedings of ACL 2005.

38. Pyatkin, V., Klein, A., Tsarfaty, R., & Dagan, I. (2020). QADiscourse: Discourse relations as QA pairs: Representation, crowdsourcing and baselines. Proceedings of EMNLP 2020.

39. Pyatkin, V., Roit, P., Michael, J., Goldberg, Y., Tsarfaty, R., & Dagan, I. (2021). Asking it all: Generating contextualized questions for any semantic role. Proceedings of EMNLP 2021.

40. Rajpurkar, P., Zhang, J., Lopyrev, K., & Liang, P. (2016). SQuAD: 100,000+ questions for machine comprehension of text. Proceedings of EMNLP 2016.

41. Rai, S. I., Croce, D., & Basili, R. (2025). Injecting frame semantics into large language models via prompt-based fine-tuning. Proceedings of *SEM 2025.

42. Roit, P., Klein, A., Stepanov, D., Mamou, J., Michael, J., Stanovsky, G., Zettlemoyer, L., & Dagan, I. (2020). Controlled crowdsourcing for high-quality QA-SRL annotation. Proceedings of ACL 2020.

43. Shen, D., & Lapata, M. (2007). Using semantic roles to improve question answering. Proceedings of EMNLP-CoNLL 2007.

44. Shi, P., & Lin, J. (2019). Simple BERT models for relation extraction and semantic role labeling. arXiv.

45. Strubell, E., Verga, P., Belanger, D., & McCallum, A. (2018). Linguistically-informed self-attention for semantic role labeling. Proceedings of EMNLP 2018.

46. Surdeanu, M., Johansson, R., Meyers, A., Marquez, L., & Nivre, J. (2008). The CoNLL-2008 shared task on joint parsing of syntactic and semantic dependencies. Proceedings of CoNLL 2008.

47. Tan, Z., Wang, M., Xie, J., Chen, Y., & Shi, X. (2018). Deep semantic role labeling with self-attention. Proceedings of AAAI 2018.

48. Torrent, T. T., Ellsworth, M., Baker, C. F., Dzikovska, M., Scheffczyk, J., Bonial, C., & Hwang, J. D. (2022). Framenet 2022: Towards a richer, more multilingual frame-semantic resource. LREC 2022 resources track materials.

49. Wang, X., Liu, Y., Xu, Y., Zhao, W. X., & Wen, J.-R. (2024). A survey of large language models. arXiv.

50. Weiss, D. B., Roit, P., Klein, A., Ernst, O., & Dagan, I. (2021). QA-Align: Representing cross-text content overlap by aligning question-answer propositions. Proceedings of EMNLP 2021.

51. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems.

52. Xue, N., & Palmer, M. (2004). Calibrating features for semantic role labeling. Proceedings of EMNLP 2004.

53. Yih, W.-T., Chang, M.-W., He, X., & Gao, J. (2015). Semantic parsing via staged query graph generation: Question answering with knowledge base. Proceedings of ACL 2015.

54. Zhou, J., & Xu, W. (2015). End-to-end learning of semantic role labeling using recurrent neural networks. Proceedings of ACL-IJCNLP 2015.